MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1MDCECT104- Creative Robotics

Duration: 1 hour **Maximum Marks: 35**

Answer *any 35* questions

Each Question carries ONE mark

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

**Remember (K), Understand (U), Apply (A), Analyse (An)*

1	Which of the following is a microcontroller used in Arduino? a) ATmega328P b) PIC16F84A c) ARM Cortex M3 d) 8051	K	1
2	How many digital I/O pins are available on the Arduino Uno board? a) 14 b) 10 c) 12 d) 16	K	1
3	Which of the following pins is not an analog pin on the Arduino Uno? a) A0 b) A1 c) A5 d) D13	K	1
4	What is the purpose of the Vin pin on an Arduino board? a) To supply external power b) To ground the circuit c) To output voltage d) To reset	U	1
5	In Arduino, the setup() function: a) Runs once when the program starts b) Runs repeatedly c) Defines logic d) Initializes	K	1
6	Which software is used for writing Arduino programs? a) Arduino IDE b) Atmel Studio c) MPLAB d) Proteus	K	1
7	Which function is used to configure a pin as INPUT or OUTPUT? a) pinMode() b) digitalWrite() c) analogRead() d) Serial.begin()	K	2
8	Which of the following is a valid pin mode? a) INPUT_PULLDOWN b) HIGH c) OUTPUT d) LOW	K	2
9	The function digitalWrite(pin, value) is used to: a) Write analog signals b) Set a digital pin c) Read digital pin d) Start serial	U	2
10	Which pin would you use with digitalWrite() to turn on an LED on pin 13? a) digitalWrite(13, HIGH) b) digitalWrite(A0, HIGH) c) digitalWrite(0, HIGH) d) A5	U	2
11	What is the maximum value for analogWrite() on a PWM pin? a) 1023 b) 255 c) 4096 d) 128	K	2
12	Which pins on the Arduino Uno support PWM? a) 3, 5, 6, 9, 10, 11 b) 0, 1, 2, 3 c) A0-A3 d) 2, 4, 7, 8	K	2
13	Which function is used to read an analog input? a) digitalRead() b) analogWrite() c) analogRead() d) digitalWrite()	K	2
14	The voltage divider formula is used to: a) Divide the input voltage b) Multiply the output voltage c) Measure current d) Read	U	2
15	Which function initializes serial communication in Arduino? a) Serial.begin() b) Serial.print() c) Serial.read() d) Serial.end()	K	2
16	What is the default baud rate for Serial.begin()? a) 4800 b) 9600 c) 115200 d) 14400	K	2
17	What does a FOR loop do in Arduino? a) Repeats code b) Executes once c) Waits for input d) Skips program	U	1
18	In a FOR loop, which is the correct syntax to increment by 1? a) i-- b) i++ c) i+=2 d) i-=2	K	1

19	A WHILE loop will continue to execute until: a) The condition is false b) The condition is true c) Time passes d) 100 iterations	K	1
20	Which of the following is the correct syntax for a WHILE loop? a) while (condition) b) for (condition) c) if (condition) d)while(increment)	K	1
21	Which pin is commonly used to connect an LED on the Arduino Uno? a) A0 b) D13 c) Vin d) GND	K	1
22	What resistor value is typically used with an LED to limit current? a) 100Ω b) 220Ω c) 1kΩ d) 10Ω	U	1
23	Which function must be called to configure an LED pin as an output? a) pinMode() b) digitalWrite() c) analogRead() d) delay()	K	1
24	The purpose of the setup() function is to: a) Define all configurations b) Contain main logic c) Run in loop d) Stop the program	K	1
25	Which function is used to add a delay in Arduino? a) delay() b) wait() c) pause() d) stop()	K	1
26	How many milliseconds are in one second for the delay function? a) 100 b) 1000 c) 10 d) 500	K	1
27	Which loop can be used to blink an LED repeatedly? a) FOR loop b) IF statement c) SWITCH d) DO-WHILE	U	1
28	In a FOR loop, how can you control the delay between LED blinks? a) Using delay() b) Using pinMode() c) Using Serial.print() d) Without delay	U	1
29	What is the purpose of an ultrasonic sensor in Arduino projects? a) Measure temperature b) Detect light c) Measure distance d) Detect sound	K	2
30	Which pins are used to connect an ultrasonic sensor to an Arduino? a) Trig & Echo b) GND & VCC c) SDA & SCL d) RX & TX	U	2
31	Which sensor is used to detect flame in Arduino projects? a) IR sensor b) Ultrasonic sensor c) Flame sensor d) Temperature sensor	K	2
32	The MQ2 sensor is used to detect: a) Smoke b) Temperature c) Humidity d) Sound	K	2
33	What does an LDR (Light Dependent Resistor) do? a) Measures light intensity b) Measures distance c) Detects temperature d) Reads sound	U	2
34	What happens to an LDR's resistance as light intensity increases? a) Increases b) Decreases c) Remains constant d) Fluctuates	U	2
35	Which motor type is used in a simple robotic arm? a) Servo motor b) Stepper motor c) DC motor d) Brushless motor	K	2
36	Which signal is used to control a servo motor in Arduino? a) PWM b) Analog c) Digital d) Serial	U	2
37	What is the purpose of a DC motor driver module in Arduino projects? a) Control DC motors b) Control stepper motors c) Control servo motors d) Control LEDs	U	2
38	Which pins are commonly used to connect a DC motor to a motor driver module? a) IN1, IN2 b) VCC, GND c) SDA, SCL d) Trig, Echo	U	2
39	What is the primary function of a geared DC motor? a) Increase torque b) Increase speed c) Increase resistance d) Decrease current	U	2
40	Which command can stop the motion of a DC motor connected to a driver module? a) digitalWrite(pin, LOW) b) analogWrite(pin, HIGH) c) Serial.end() d) pinMode(pin, INPUT)	U	2

1 * 35 = 35